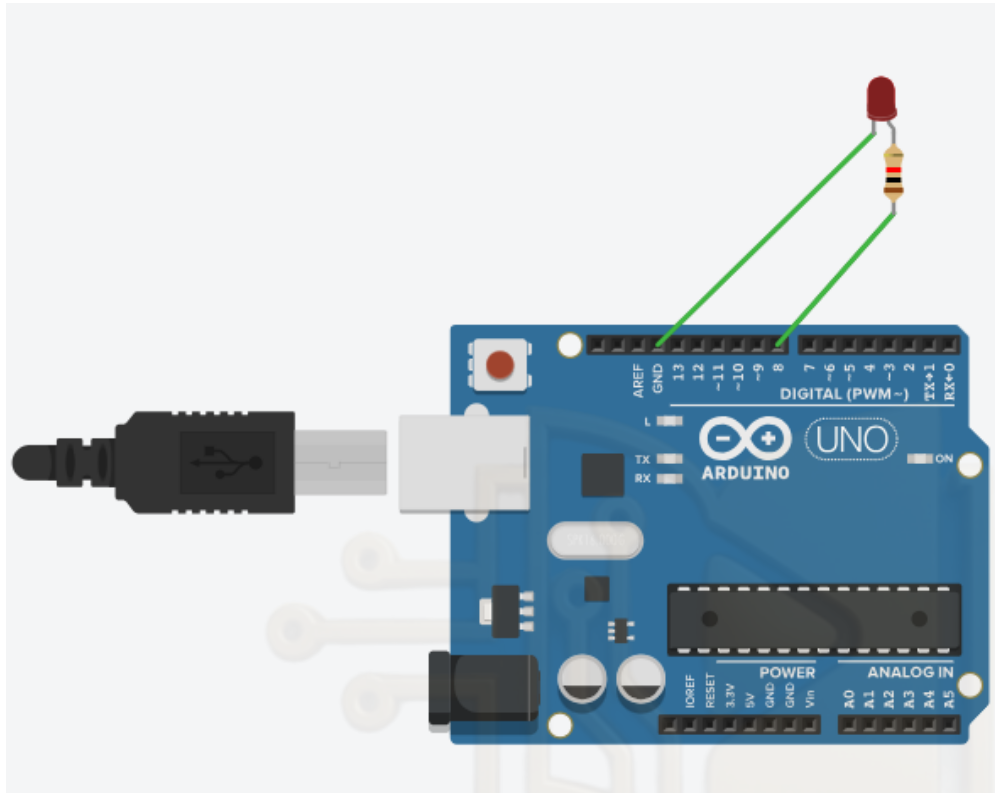


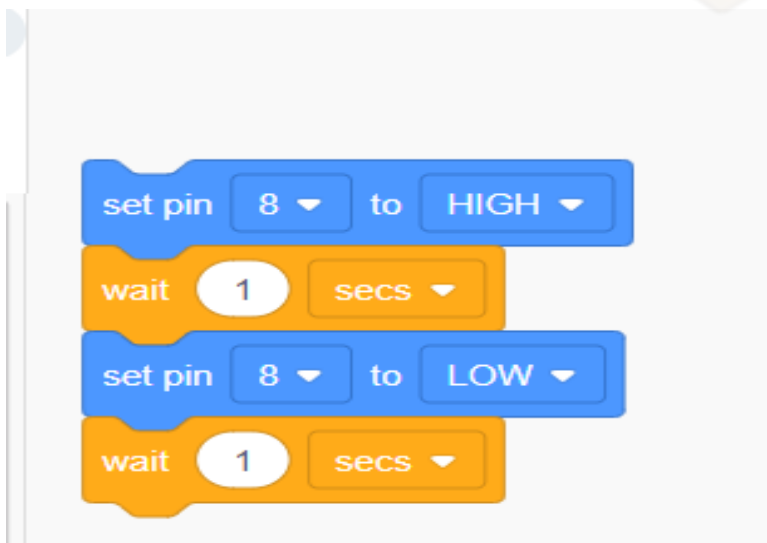
IoT projects:

1. Simple LED on and Off:

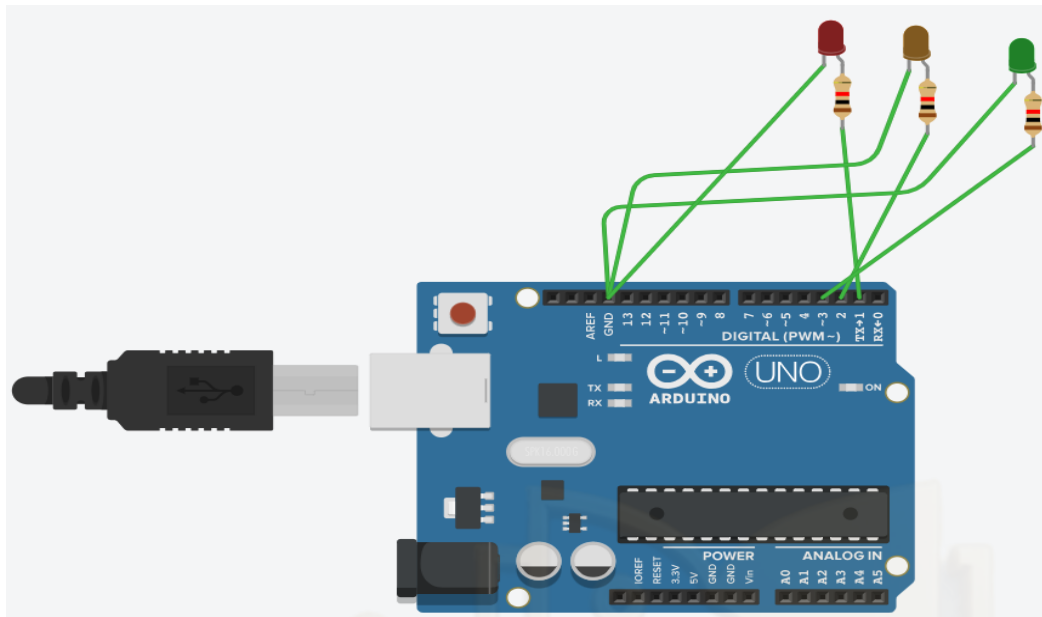
Circuit Configuration:



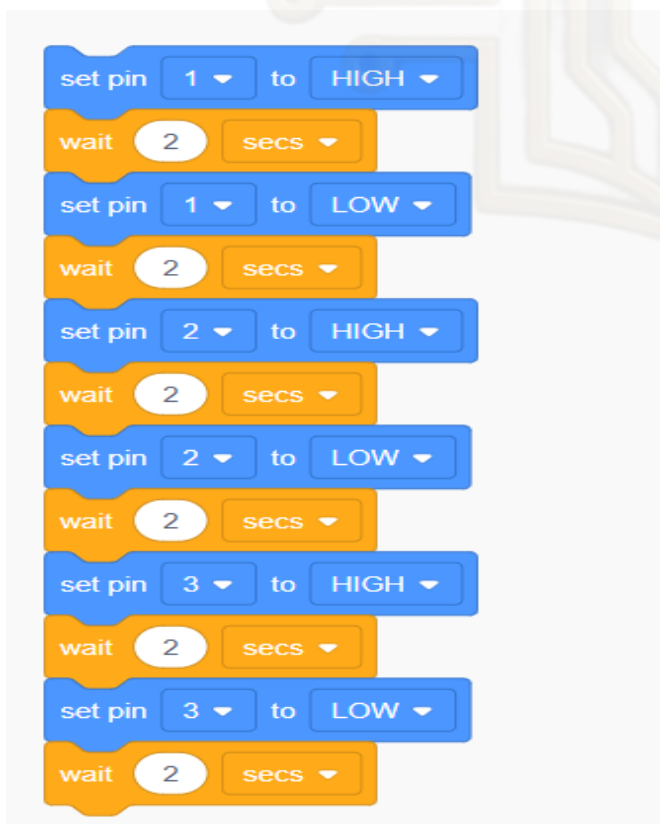
Code:



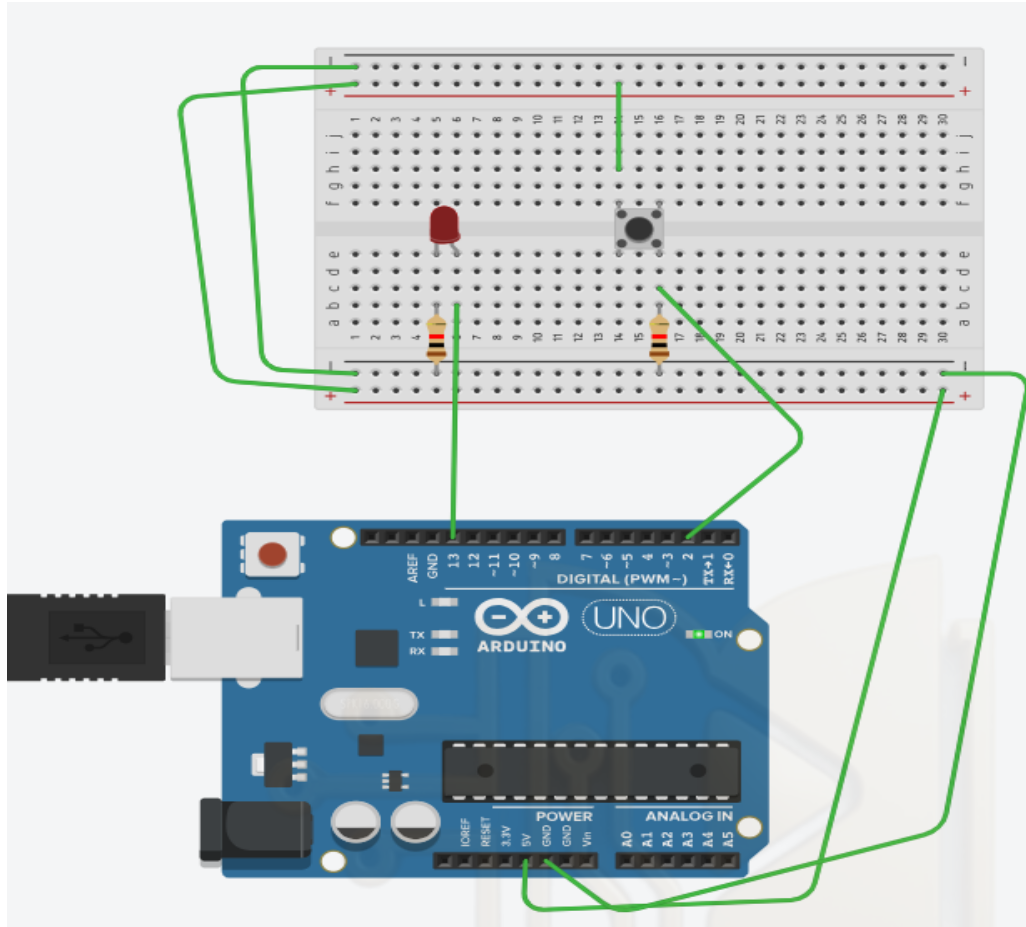
2. 3 Led RGB with a delay of 2sec:



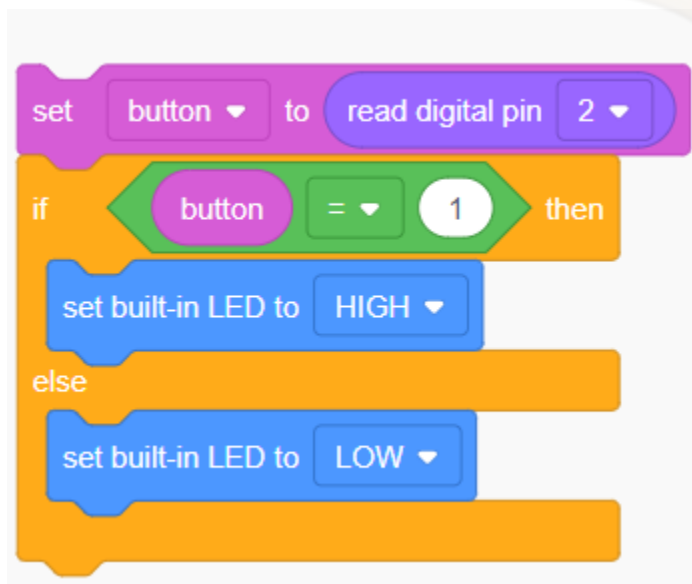
Code:



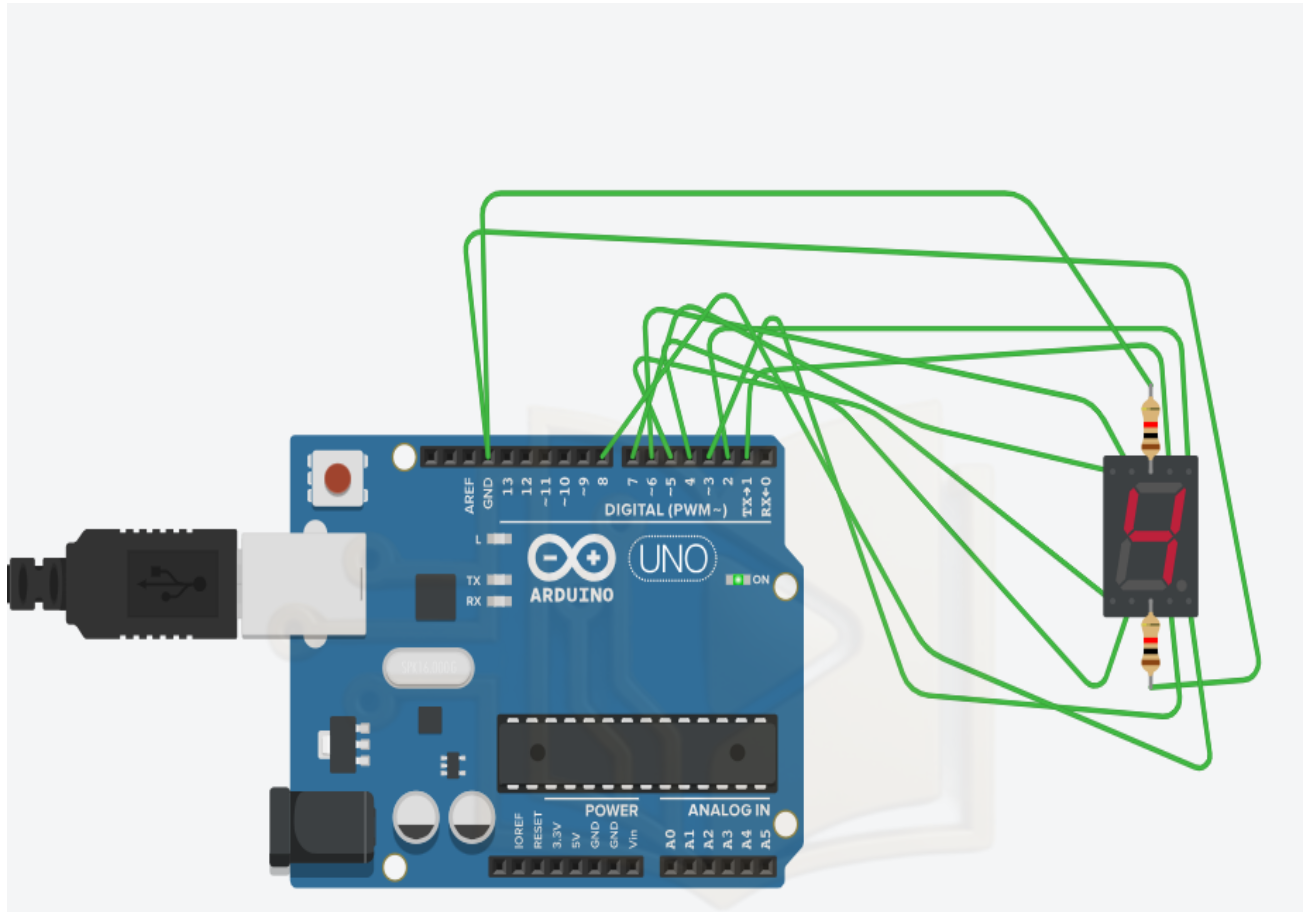
3. Led On and Off with push button:



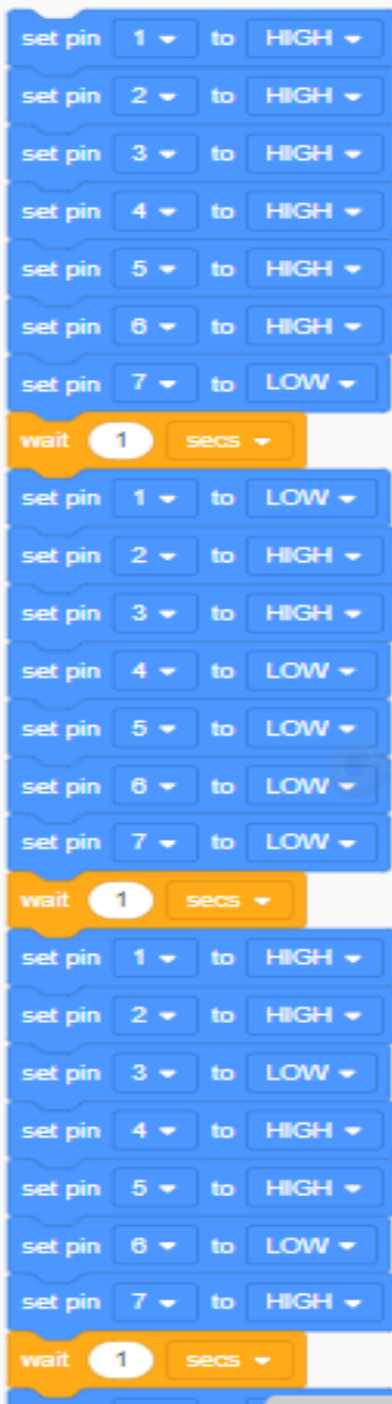
Code:



4. 7-Segment Display:

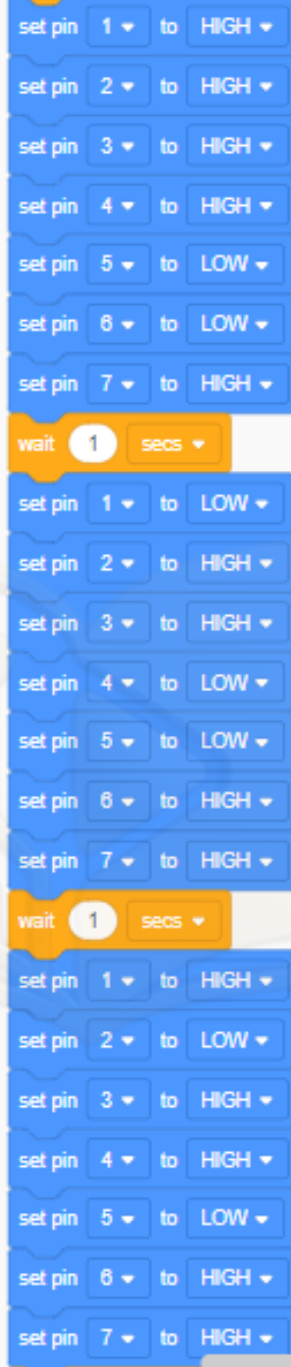


Code:



Scratch code for a 7-segment display showing the digit 1. The code consists of three identical blocks, each containing seven 'set pin' blocks followed by a 'wait 1 secs' block. The pin states for each block are: pin 1 to HIGH, pin 2 to HIGH, pin 3 to HIGH, pin 4 to LOW, pin 5 to LOW, pin 6 to LOW, and pin 7 to LOW.

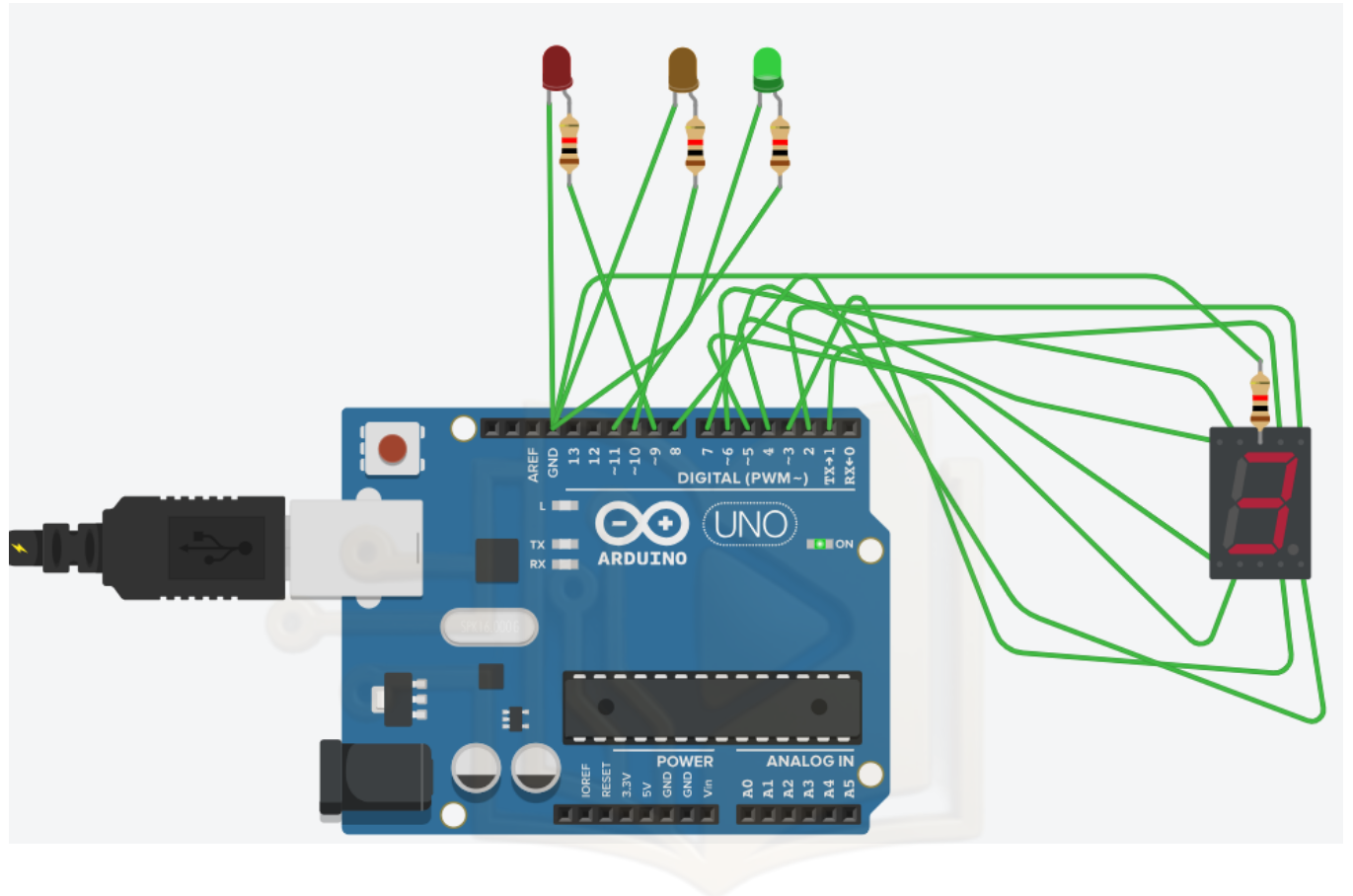
```
set pin 1 to HIGH
set pin 2 to HIGH
set pin 3 to HIGH
set pin 4 to LOW
set pin 5 to LOW
set pin 6 to LOW
set pin 7 to LOW
wait 1 secs
set pin 1 to LOW
set pin 2 to HIGH
set pin 3 to HIGH
set pin 4 to LOW
set pin 5 to LOW
set pin 6 to LOW
set pin 7 to LOW
wait 1 secs
set pin 1 to HIGH
set pin 2 to HIGH
set pin 3 to LOW
set pin 4 to HIGH
set pin 5 to HIGH
set pin 6 to LOW
set pin 7 to HIGH
wait 1 secs
```



Scratch code for a 7-segment display showing the digit 4. The code consists of three identical blocks, each containing seven 'set pin' blocks followed by a 'wait 1 secs' block. The pin states for each block are: pin 1 to HIGH, pin 2 to HIGH, pin 3 to HIGH, pin 4 to HIGH, pin 5 to LOW, pin 6 to HIGH, and pin 7 to HIGH.

```
set pin 1 to HIGH
set pin 2 to HIGH
set pin 3 to HIGH
set pin 4 to HIGH
set pin 5 to LOW
set pin 6 to HIGH
set pin 7 to HIGH
wait 1 secs
set pin 1 to LOW
set pin 2 to HIGH
set pin 3 to HIGH
set pin 4 to LOW
set pin 5 to LOW
set pin 6 to HIGH
set pin 7 to HIGH
wait 1 secs
set pin 1 to HIGH
set pin 2 to LOW
set pin 3 to HIGH
set pin 4 to HIGH
set pin 5 to LOW
set pin 6 to HIGH
set pin 7 to HIGH
wait 1 secs
```

5. Seven-segment display with LED traffic light:



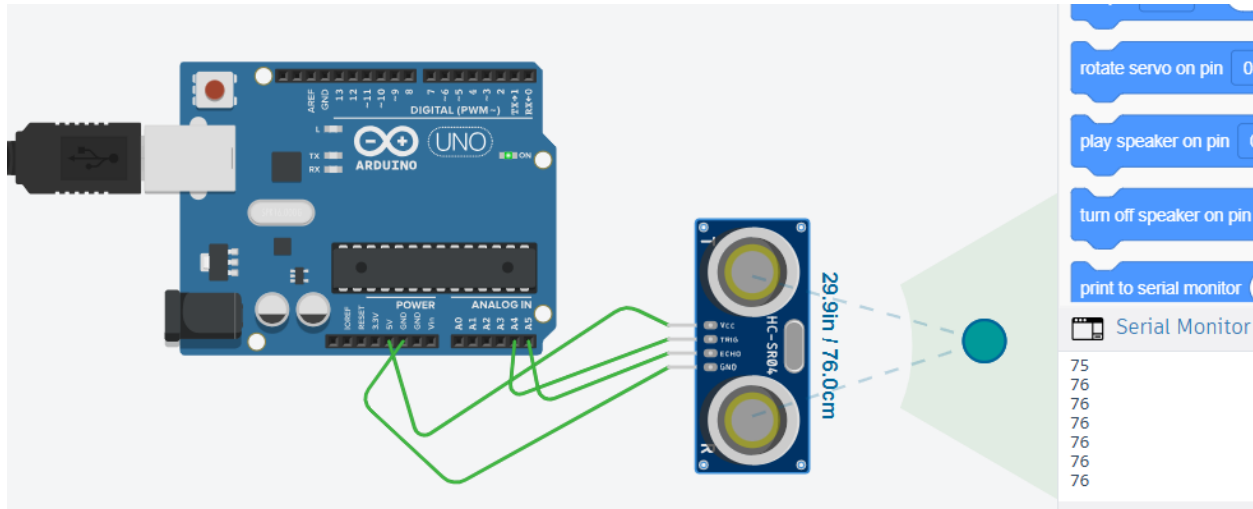
Code:

```

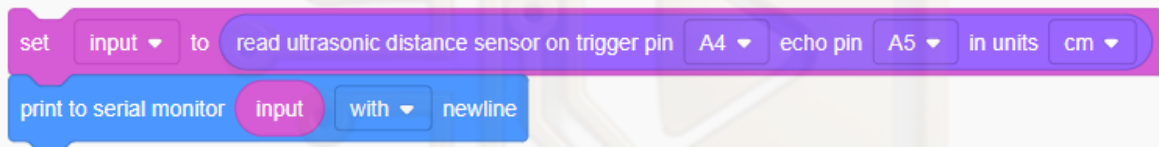
1 const int redLED    = 9;
2 const int yellowLED = 10;
3 const int greenLED  = 11;
4 const int segPins[7] = {1, 2, 3, 4, 5, 6, 7};
5 const byte digits[10][7] = {
6     {1,1,1,1,1,1,0}, // 0
7     {0,1,1,0,0,0,0}, // 1
8     {1,1,0,1,1,0,1}, // 2
9     {1,1,1,1,0,0,1}, // 3
10    {0,1,1,0,0,1,1}, // 4
11    {1,0,1,1,0,1,1}, // 5
12    {1,0,1,1,1,1,1}, // 6
13    {1,1,1,0,0,0,0}, // 7
14    {1,1,1,1,1,1,1}, // 8
15    {1,1,1,1,0,1,1}  // 9
16 };
17 void displayDigit(int num) {
18     for (int i = 0; i < 7; i++) {
19         digitalWrite(segPins[i], digits[num][i]);
20     }
21 }
22 void setup() {
23     pinMode(redLED, OUTPUT);
24     pinMode(yellowLED, OUTPUT);
25     pinMode(greenLED, OUTPUT);
26
27     for (int i = 0; i < 7; i++) {
28         pinMode(segPins[i], OUTPUT);
29     }
30 }
31 void loop() {
32     // GREEN light countdown (5 → 0)
33     digitalWrite(greenLED, HIGH);
34     for (int i = 5; i >= 0; i--) {
35         displayDigit(i);
36         delay(1000);
37     }
38     digitalWrite(greenLED, LOW);
39
40     // YELLOW light countdown (2 → 0)
41     digitalWrite(yellowLED, HIGH);
42     for (int i = 2; i >= 0; i--) {
43         displayDigit(i);
44         delay(1000);
45     }
46     digitalWrite(yellowLED, LOW);
47
48     // RED light countdown (5 → 0)
49     digitalWrite(redLED, HIGH);
50     for (int i = 5; i >= 0; i--) {
51         displayDigit(i);
52         delay(1000);
53     }
54     digitalWrite(redLED, LOW);
55 }

```

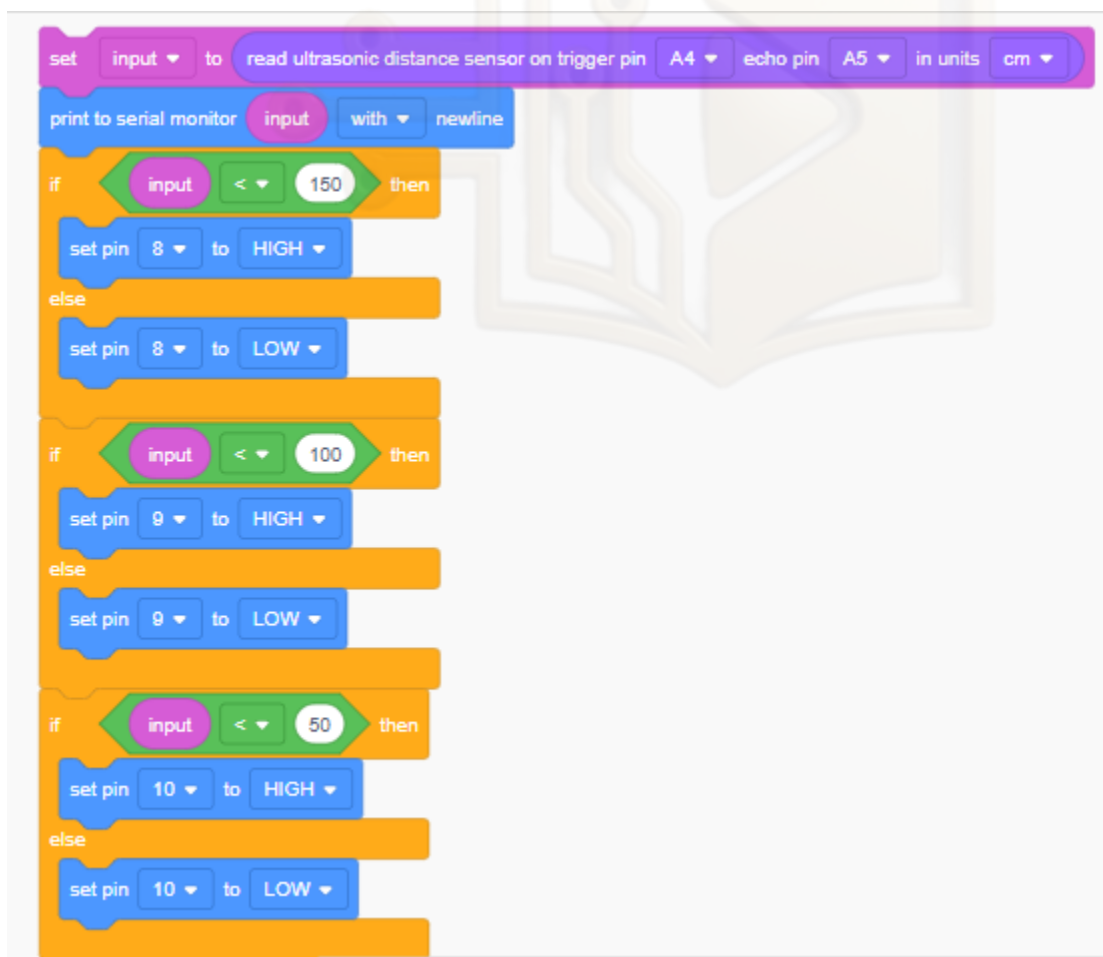
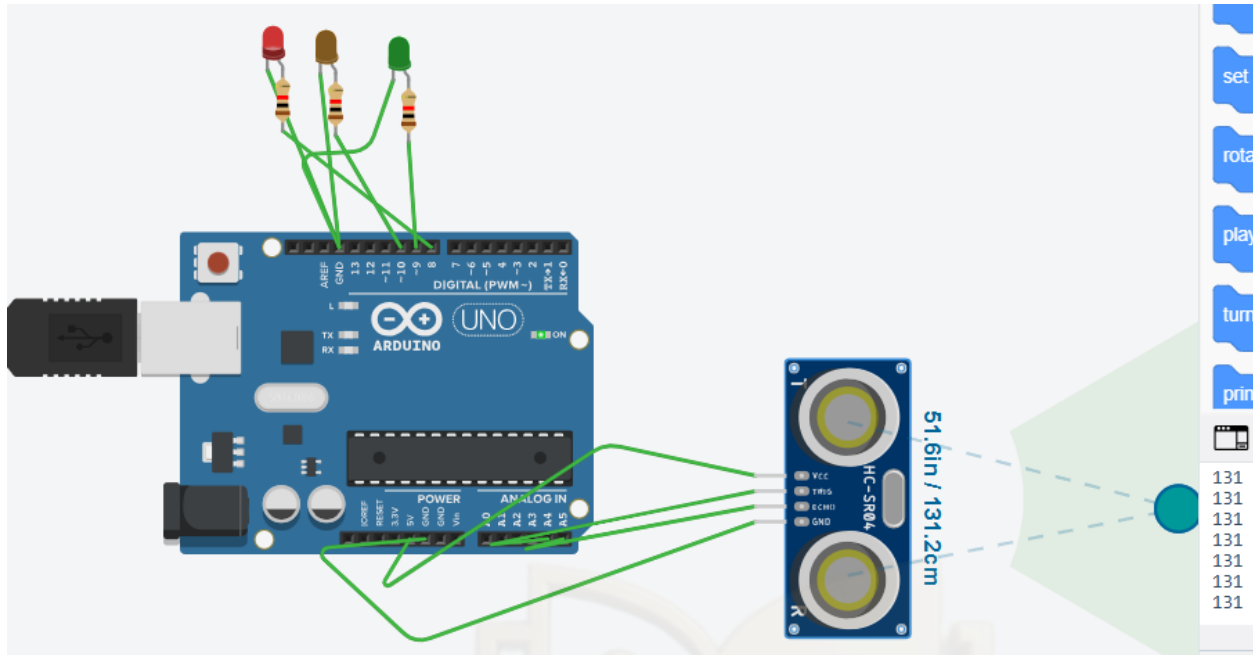
6. Ultrasonic Sensor Interfacing:



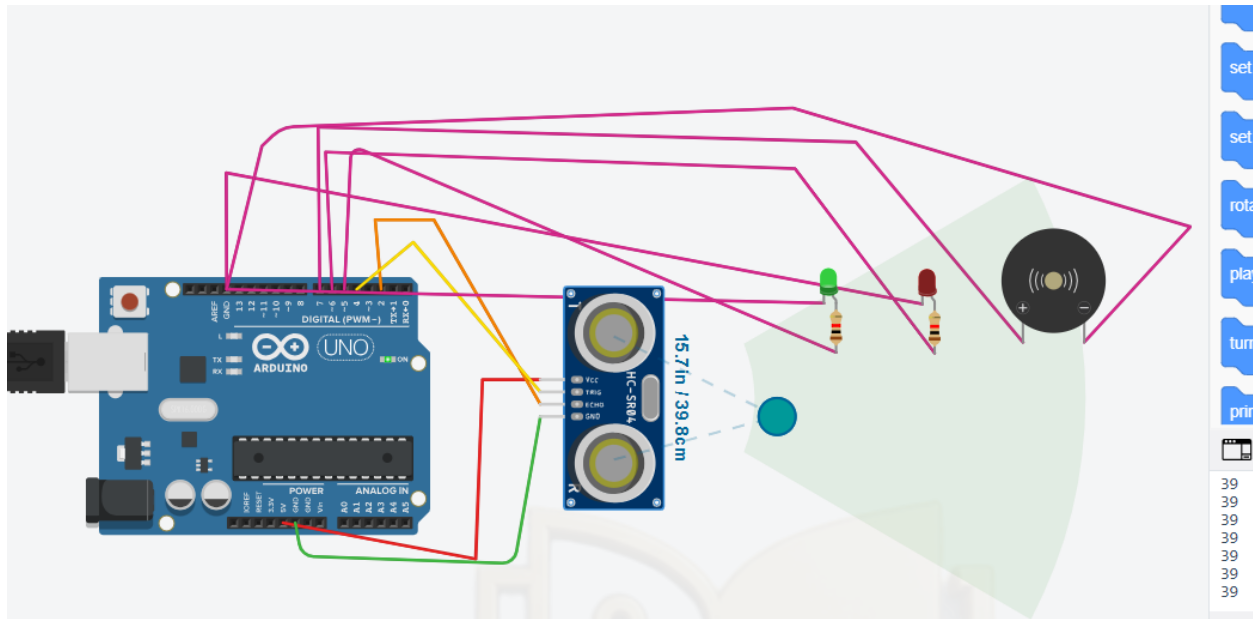
Code:



7. Distance Measurement with LED & Arduino and Ultrasonic sensors:



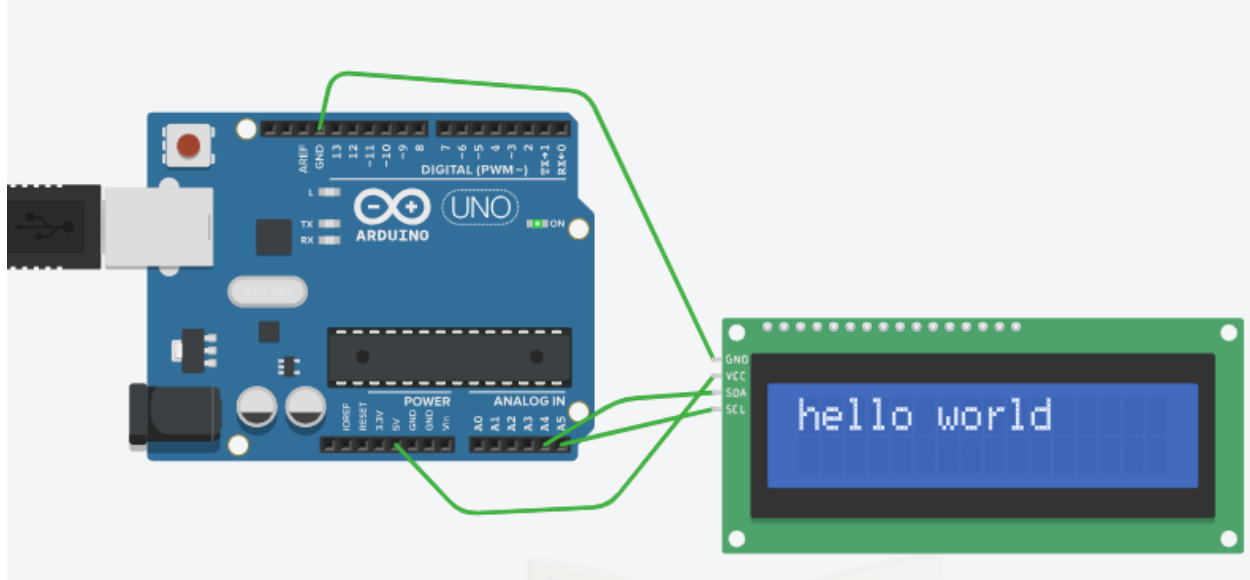
8. Distance Measurement with LED & Arduino and Ultrasonic sensors with an alarm as well:



Code:

```
set sense to read ultrasonic distance sensor on trigger pin 4 echo pin 2 in units cm
print to serial monitor sense with newline
if sense < 50 then
  set pin 5 to HIGH
  set pin 7 to HIGH
  set pin 6 to LOW
  play speaker on pin 7 with tone 60 for 1 sec
else
  set pin 6 to HIGH
  set pin 5 to LOW
```

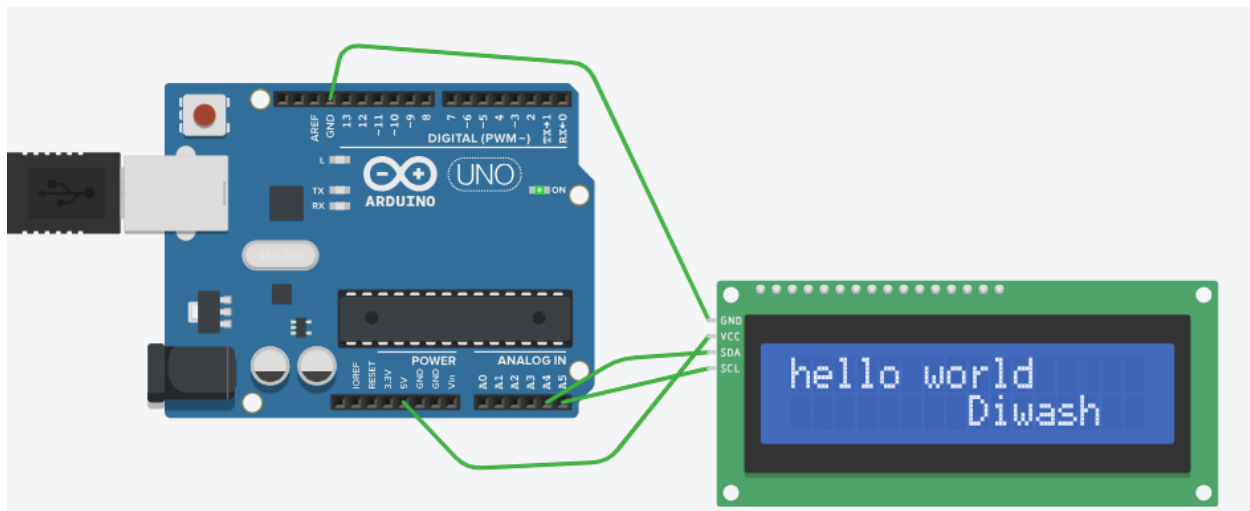
9. Print Hello world



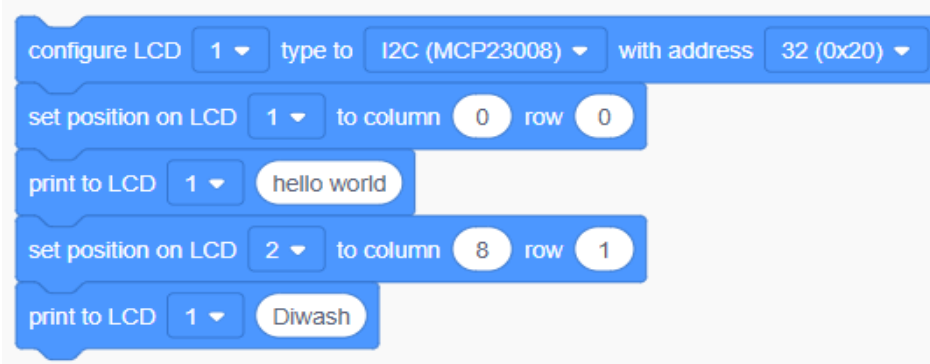
Code:



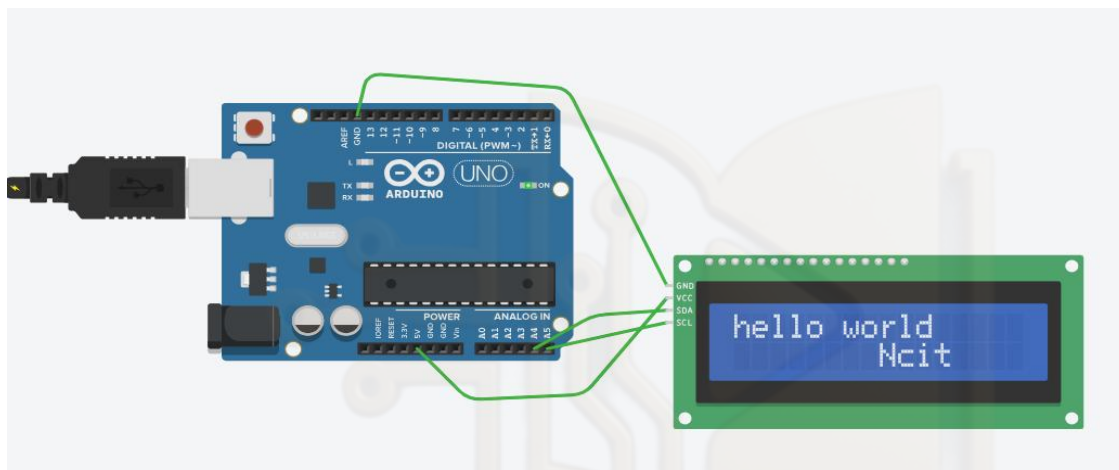
Print name:



Code:



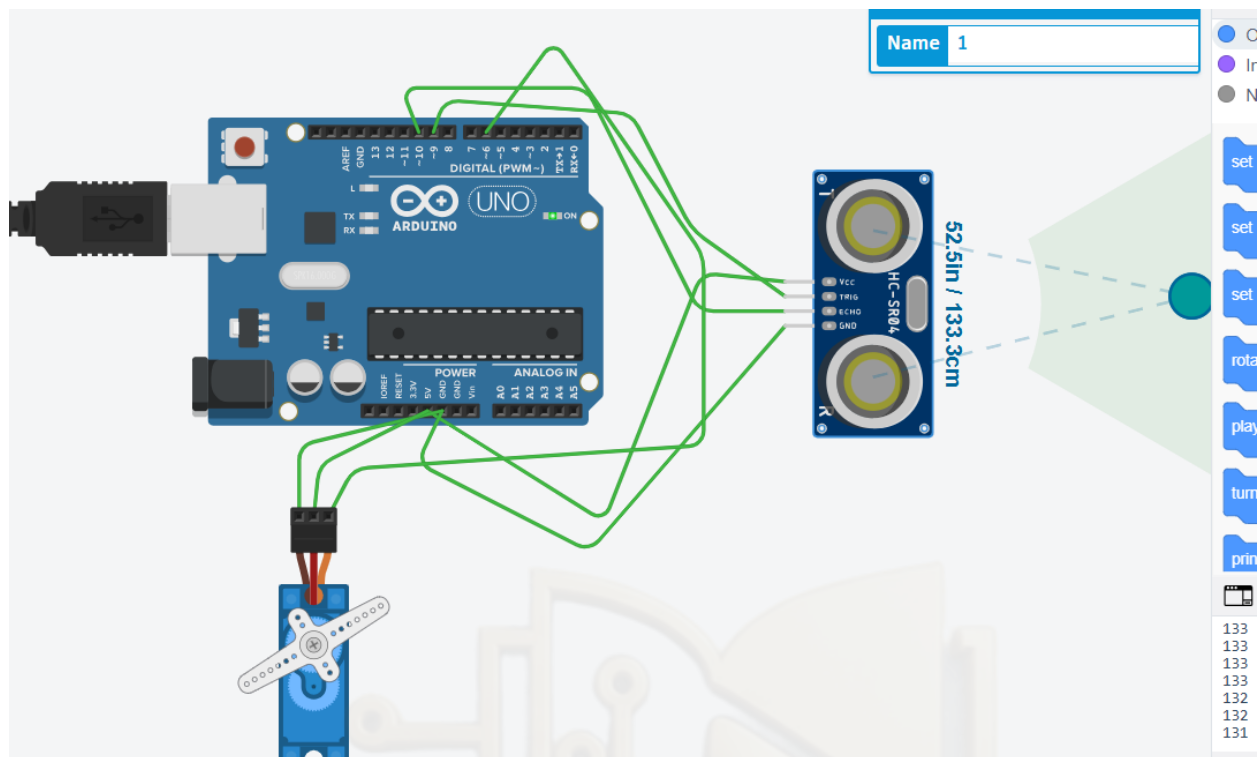
Print ncit:



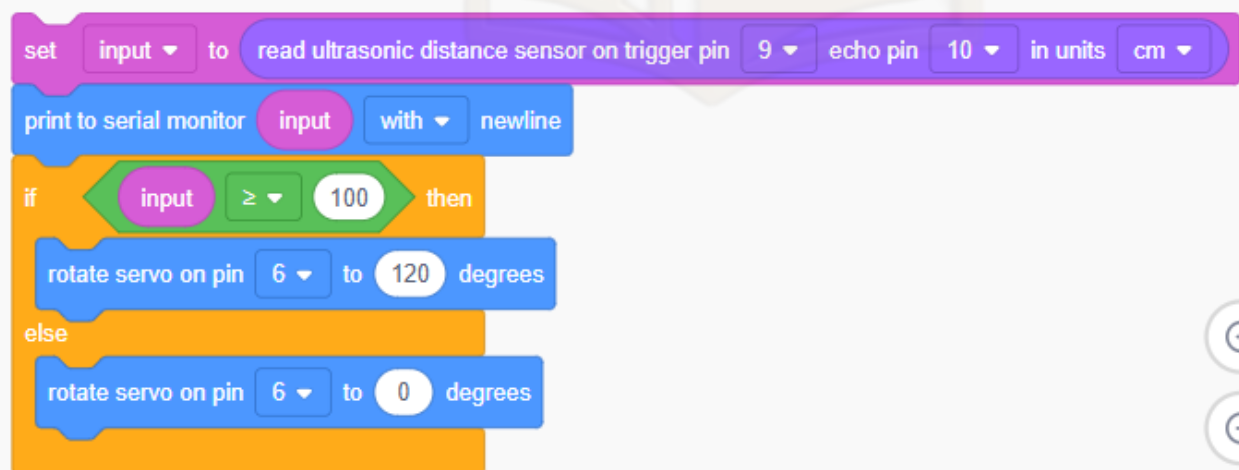
Code:



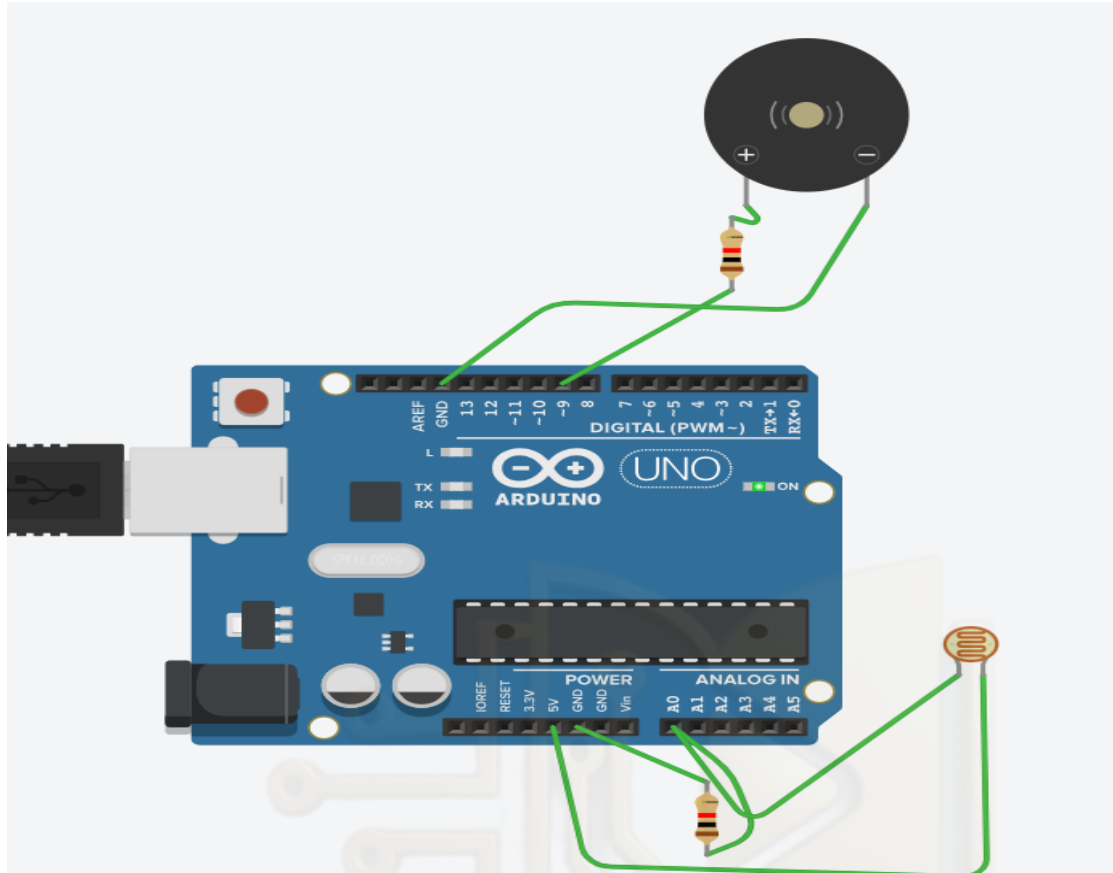
10. Ultrasonic sensor with servo motor:



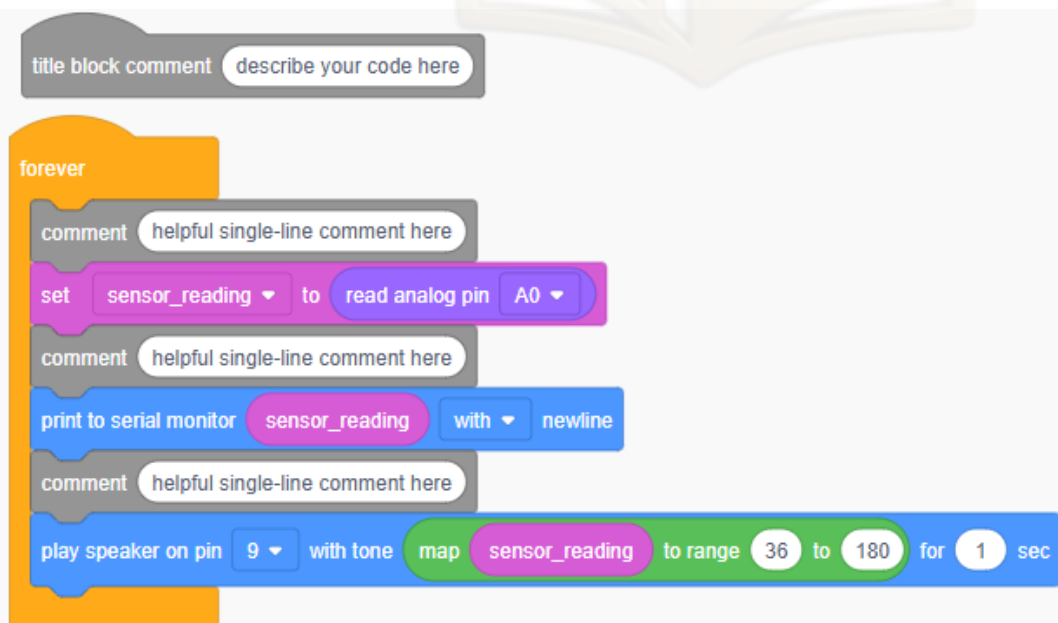
Code:



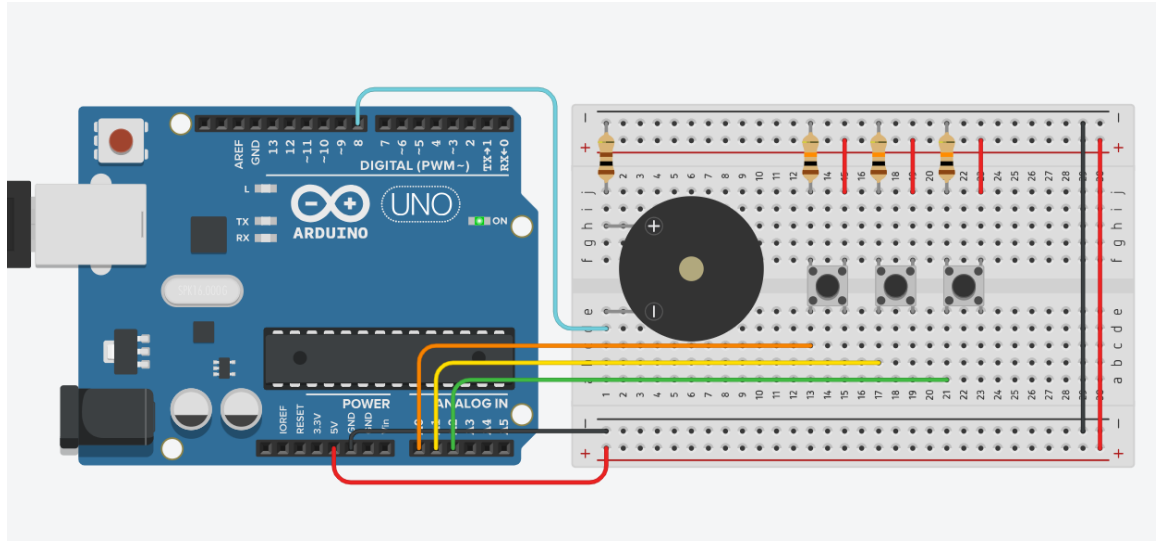
11. Tone Pitch follower:



Code:



12. Tone keyboard:



Code:



Text



1 (Arduino Uno R3

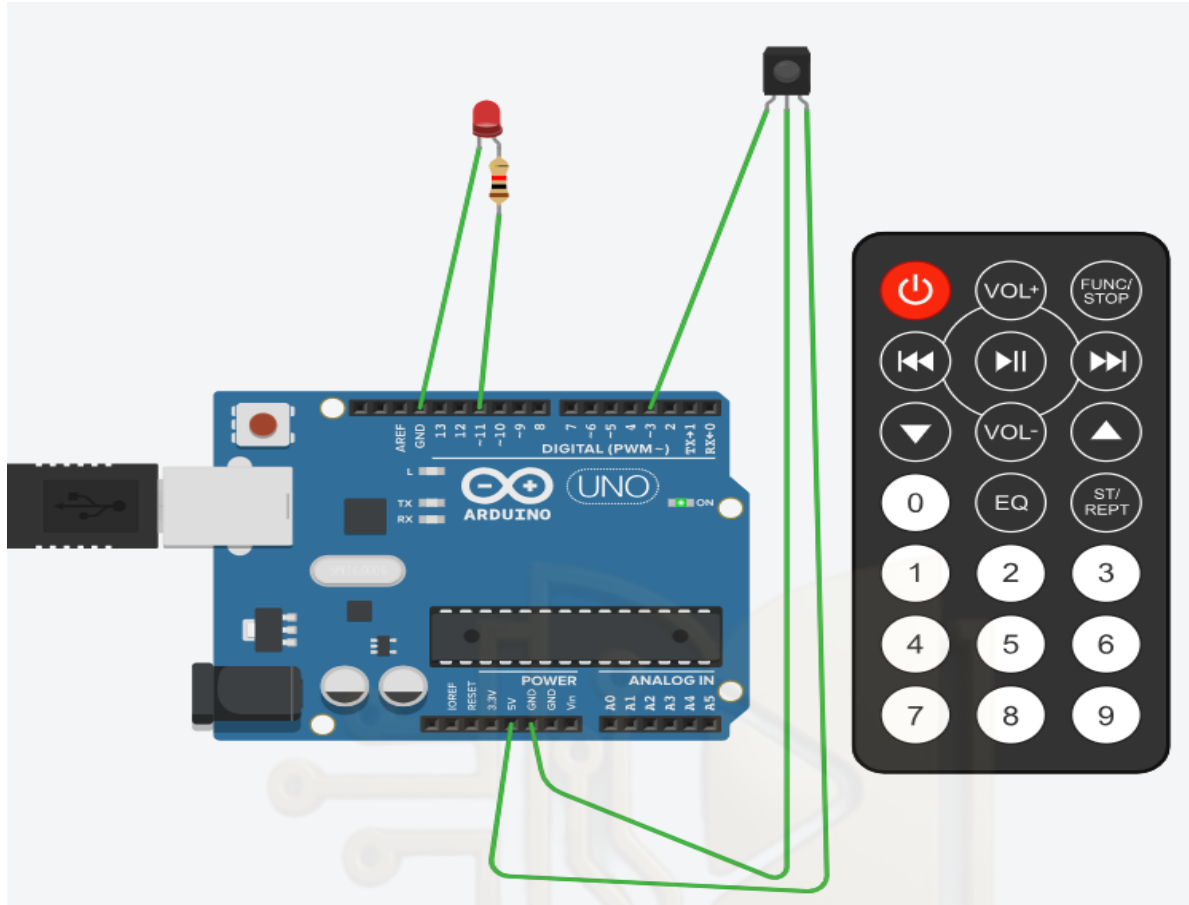
```
1  const int gasPin = A0;
2  const int tempPin = A1;
3  const int buzzer = 7;
4  const int led = 13;
5  int gasThreshold = 300;
6  float tempThreshold = 50.0;
7  void setup() {
8      pinMode(buzzer, OUTPUT);
9      pinMode(led, OUTPUT);
10     Serial.begin(9600);
11     Serial.println("Fire Alarm System Ready");
12 }
13 void loop() {
14     int gasValue = analogRead(gasPin);
15     int tempValue = analogRead(tempPin);
16     float voltage = tempValue * (5.0 / 1023.0);
17     float tempC = (voltage - 0.5) * 100.0;
18     Serial.print("Gas Value = ");
19     Serial.print(gasValue);
20     Serial.print(" | Temp = ");
21     Serial.print(tempC);
22     Serial.println(" °C");
23     if (gasValue > gasThreshold || tempC > tempThreshold) {
24         digitalWrite(led, HIGH);
25         digitalWrite(buzzer, HIGH);
26         delay(200);
27         digitalWrite(buzzer, LOW);
28         delay(200);
29     } else {
30         digitalWrite(led, LOW);
31         digitalWrite(buzzer, LOW);
32     }
33     delay(500);
34 }
```



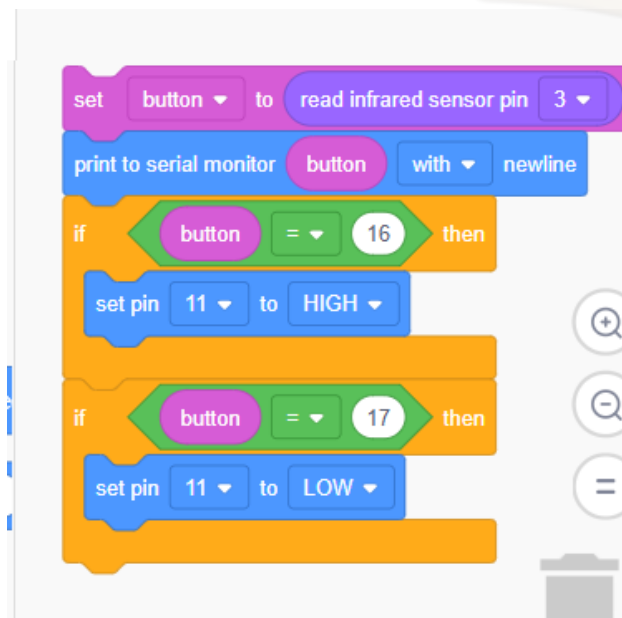
Serial Monitor

Gas Value = 307		Temp = 24.78 °C
Gas Value = 307		Temp = 24.78 °C
Gas Value = 307		Temp = 24.78 °C
Gas Value = 307		Temp = 24.78 °C
Gas Value = 307		Temp = 24.78 °C
Gas Value = 307		Temp = 24.78 °C
Gas Value = 307		Temp = 24.78 °C

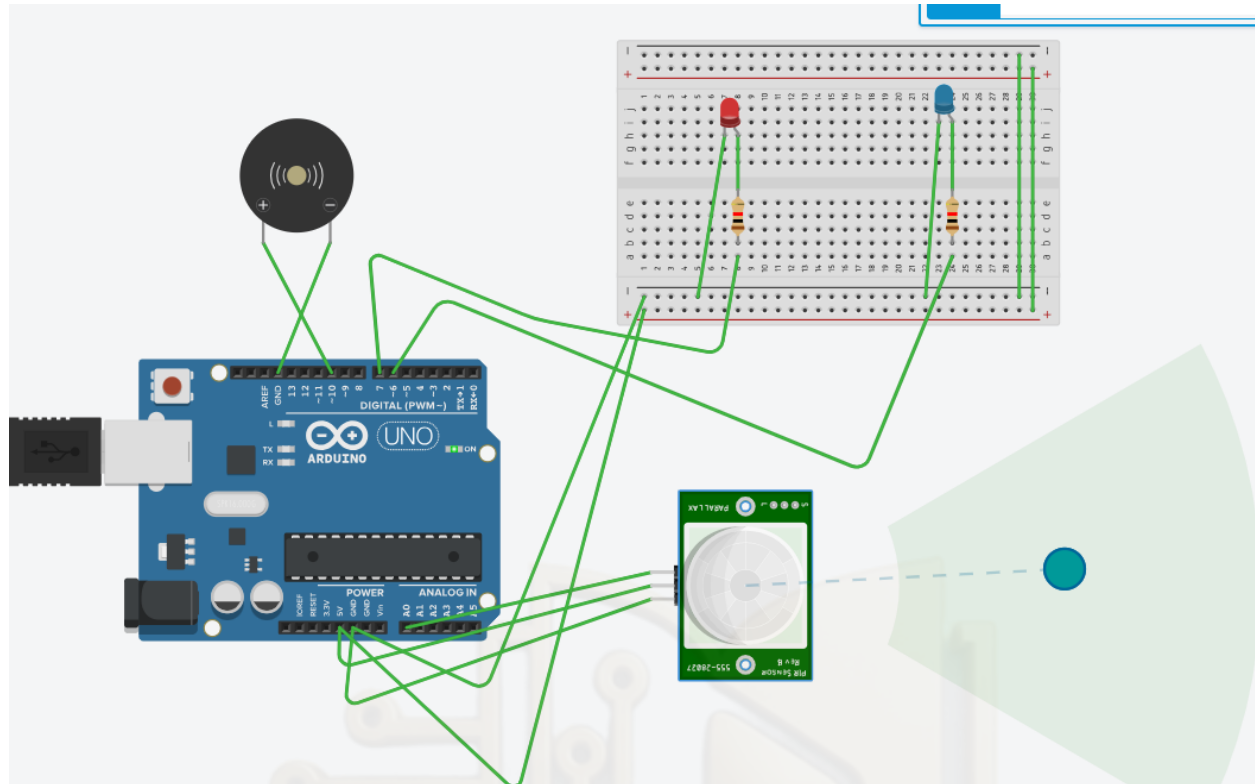
14. Controlling LEDs using an IR Remote:



Code:



15. PIR motion sensor:

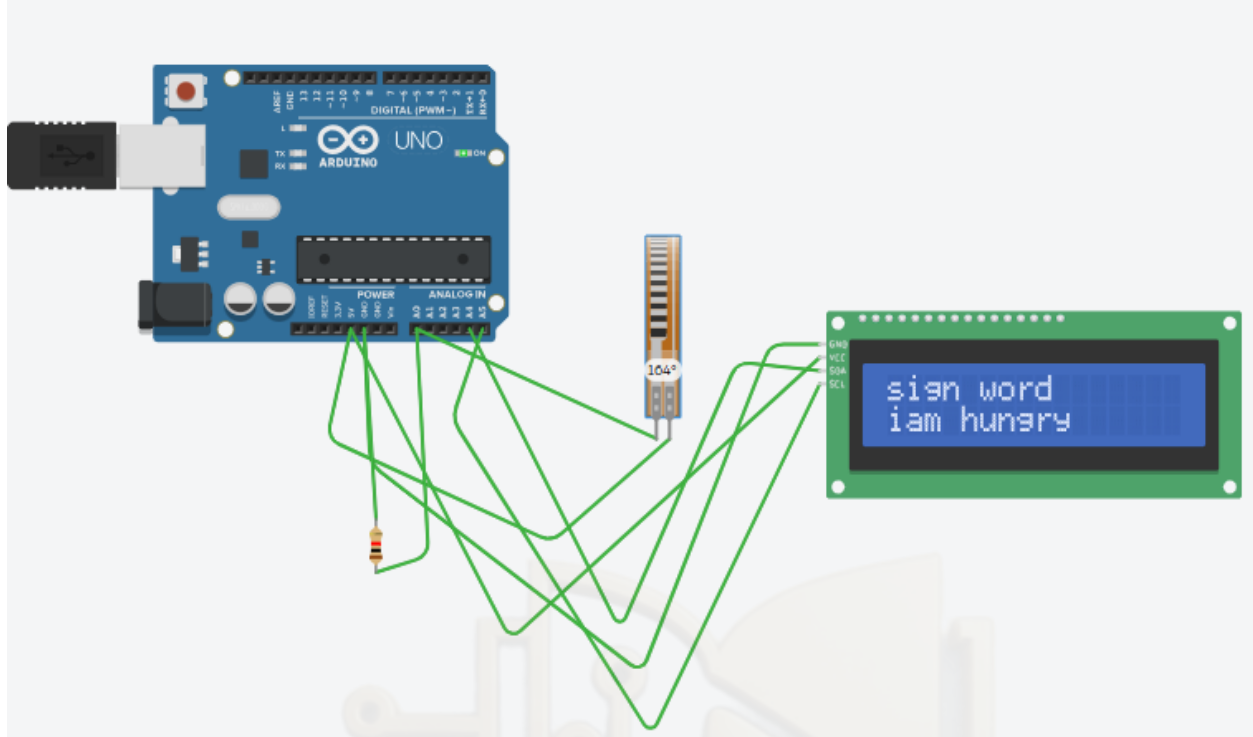


Code:

```
set sensor to read analog pin A0
print to serial monitor sensor with newline

if sensor ≥ 100 then
  set pin 7 to LOW
  set pin 6 to HIGH
  play speaker on pin 10 with tone 60 for 1 sec
else
  set pin 7 to HIGH
  set pin 6 to LOW
  set pin 10 to LOW
```

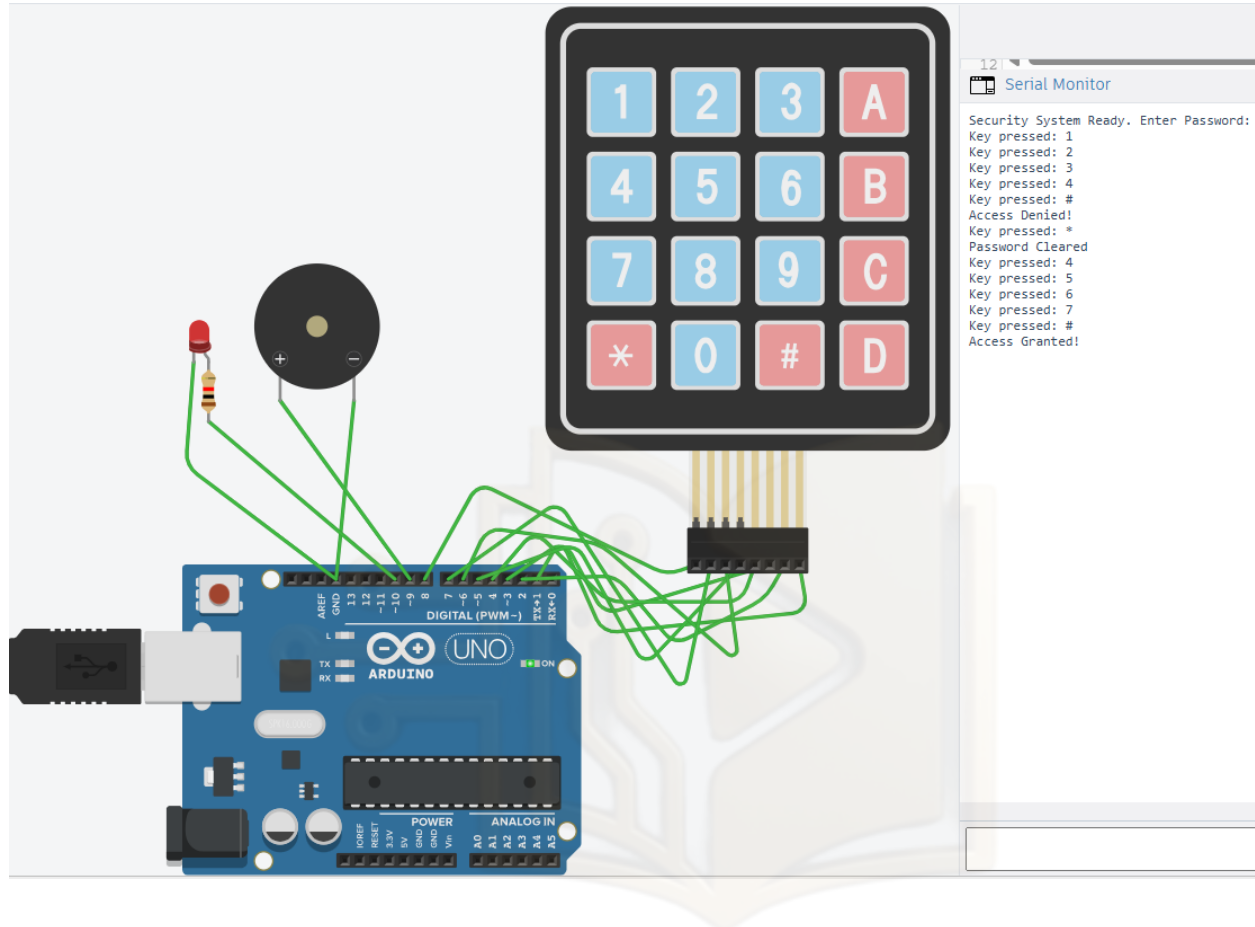
16. Flex sensor for sign language:



Code:



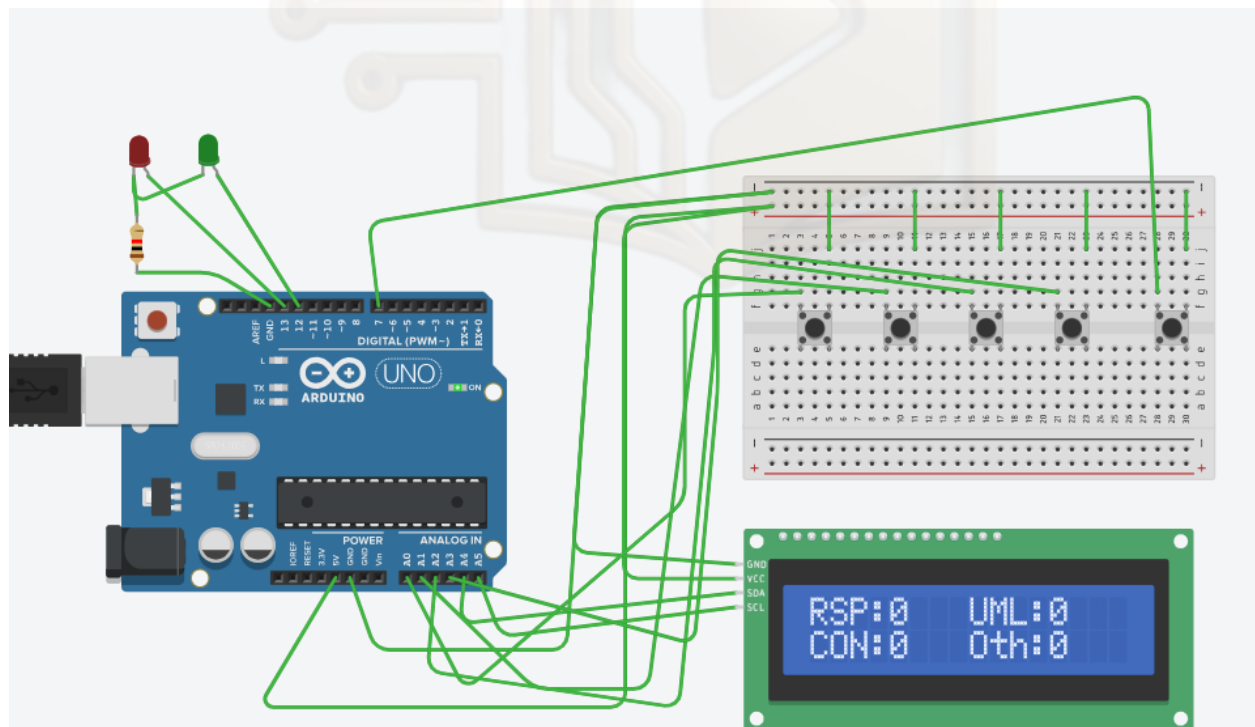
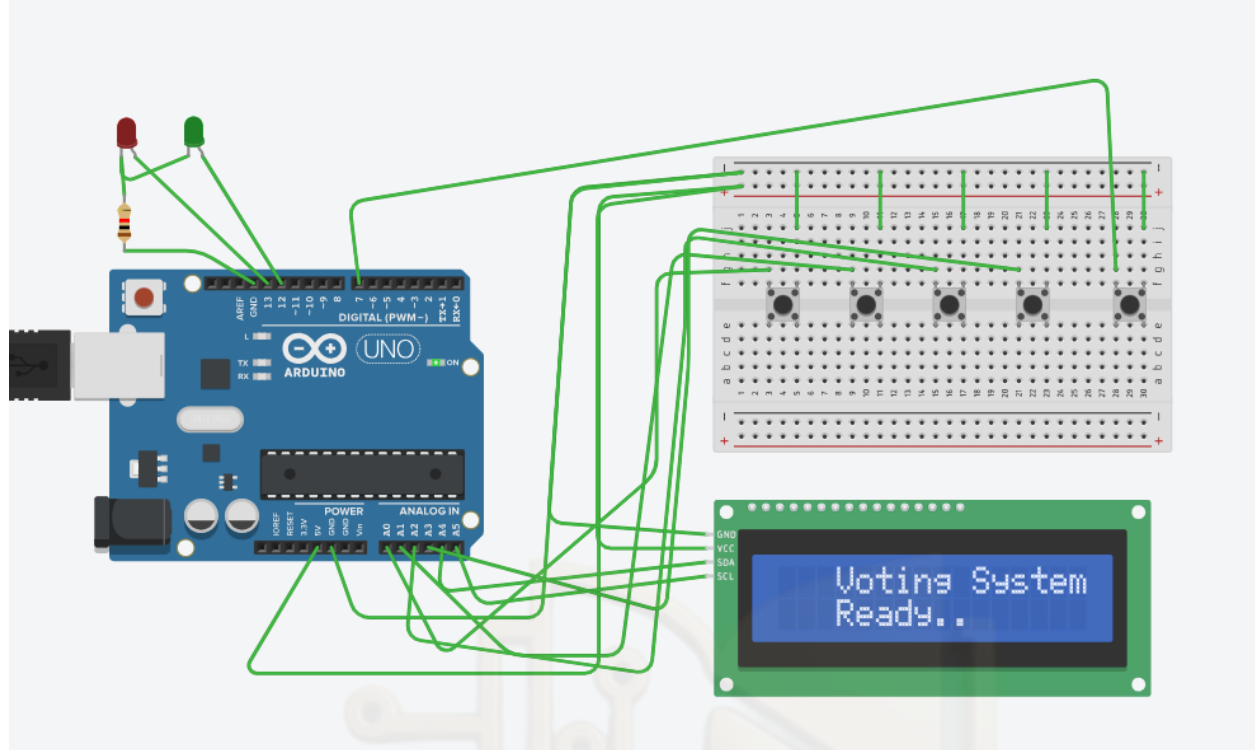
17. Password-Protected Security System project using Arduino, Keypad, Piezo Buzzer, and LED.

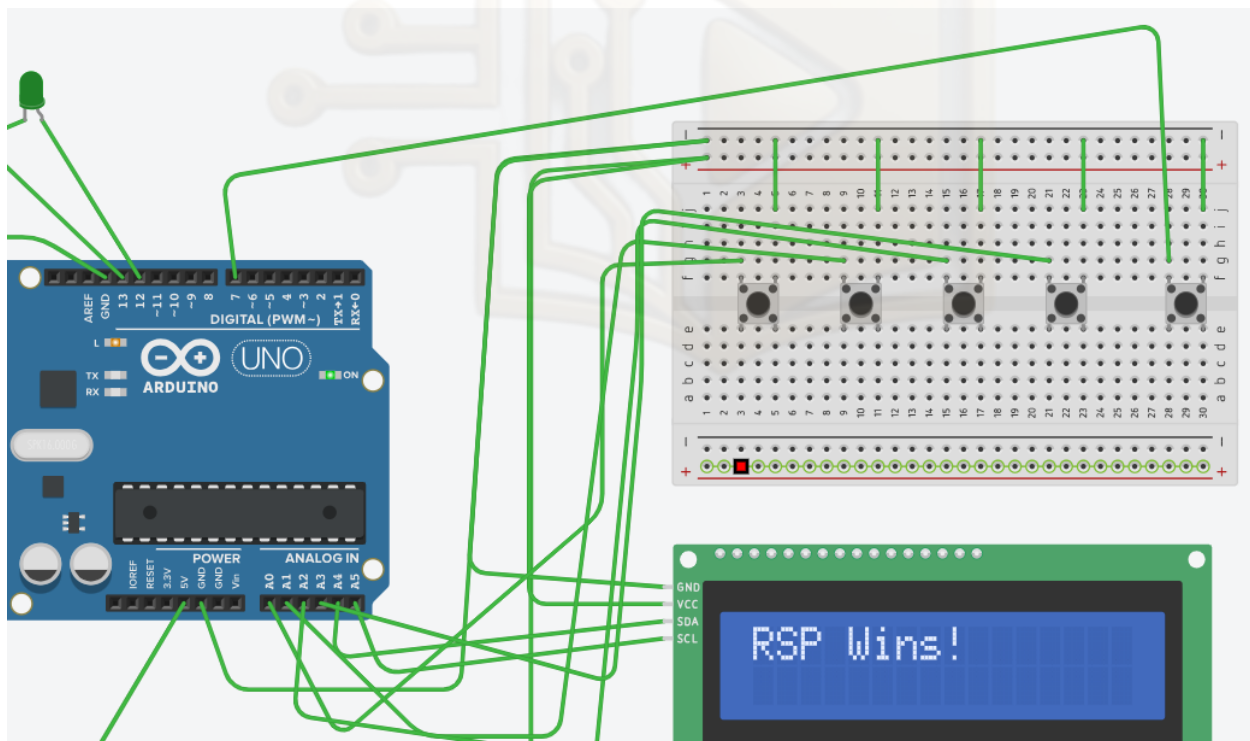
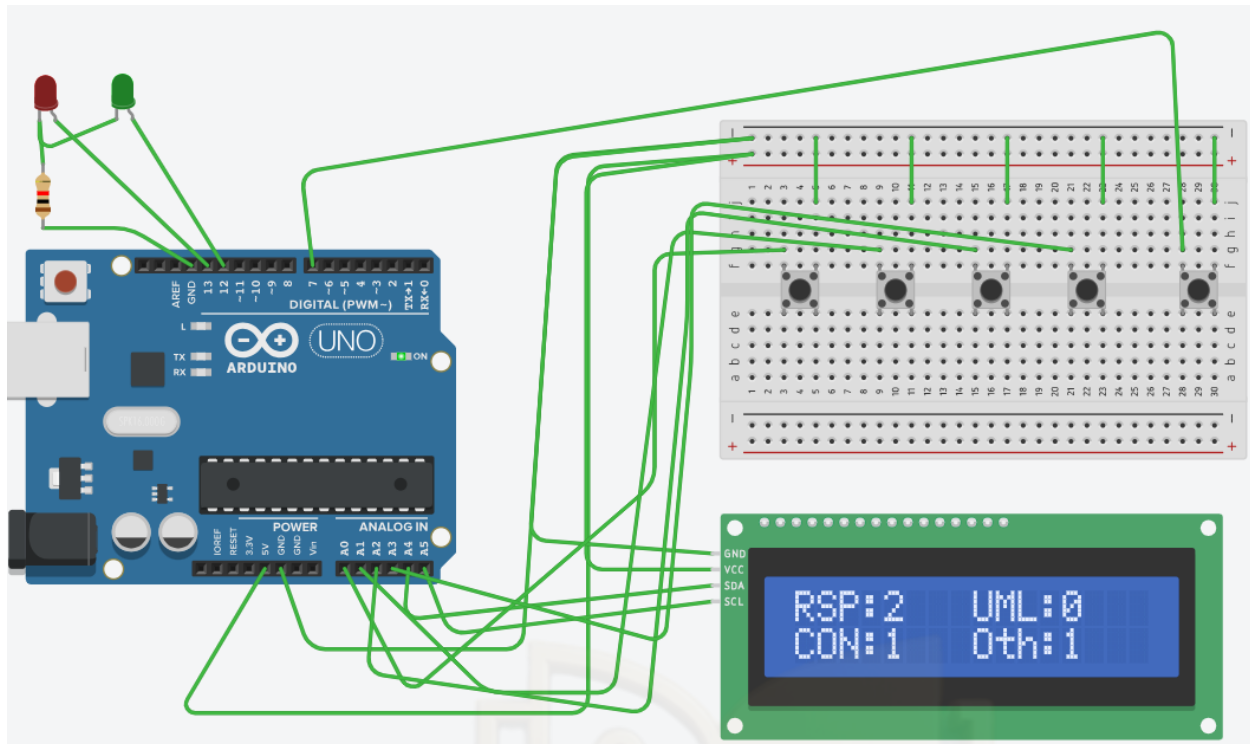


Code:

```
1 #include <Keypad.h>
2 const byte ROWS = 4;
3 const byte COLS = 4;
4 char keys[ROWS][COLS] = {
5     {'1','2','3','A'},
6     {'4','5','6','B'},
7     {'7','8','9','C'},
8     {'*','0','#','D'} };
9 byte rowPins[ROWS] = {8,7,6,5};
10 byte colPins[COLS] = {4,3,2,1};
11 Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
12 const int ledPin    = 10;
13 const int buzzerPin = 9;
14 String password      = "4567";
15 String inputPassword = "";
16 void setup() {
17     pinMode(ledPin, OUTPUT);
18     pinMode(buzzerPin, OUTPUT);
19     Serial.begin(9600);
20     Serial.println("Security System Ready. Enter Password:");
21 }
22 void loop() {
23     char key = keypad.getKey();
24
25     if (key) {
26         Serial.print("Key pressed: ");
27         Serial.println(key);
28         if (key == '#') {
29
30             if (inputPassword == password) {
31                 Serial.println("Access Granted!");
32                 digitalWrite(ledPin, HIGH);
33                 noTone(buzzerPin);
34             } else {
35                 Serial.println("Access Denied!");
36                 digitalWrite(ledPin, LOW);
37                 tone(buzzerPin, 1000, 500);
38             }
39             inputPassword = "";
40         }
41         else if (key == '*') {
42             inputPassword = "";
43             Serial.println("Password Cleared");
44         }
45         else {
46             inputPassword += key;
47         }
48     }
49 }
```

18. Voting system with LCD:





Code:

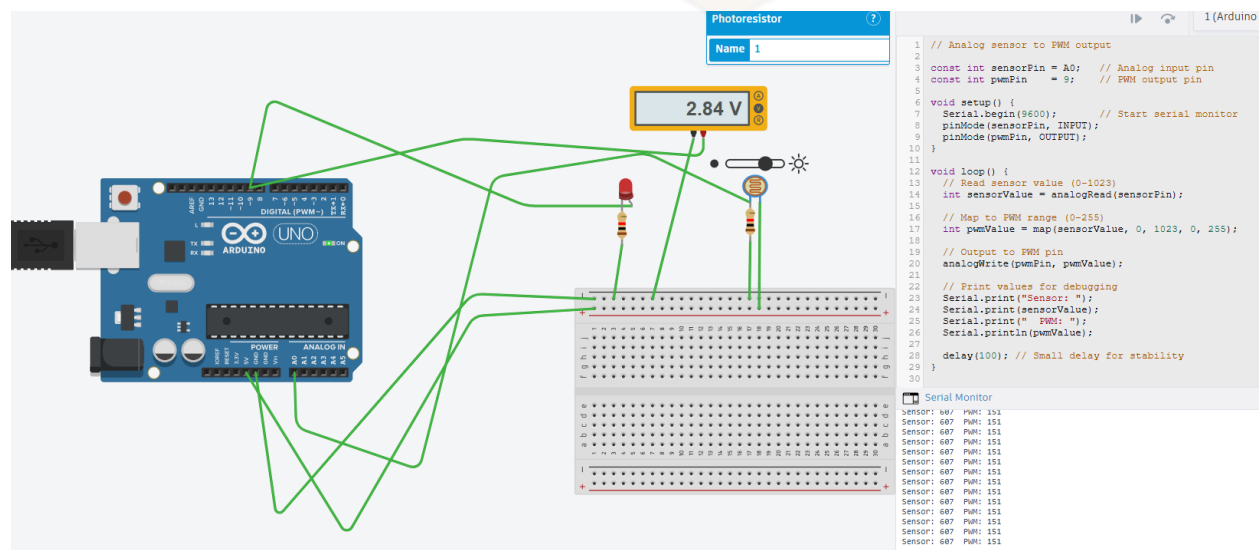
```
1  #include <Adafruit_LiquidCrystal.h>
2
3  Adafruit_LiquidCrystal lcd(0);
4  #define sw1 A0
5  #define sw2 A1
6  #define sw3 A2
7  #define sw4 A3
8  #define sw5 7    // Result button
9
10 int votes[4] = {0, 0, 0, 0};
11 void setup() {
12     pinMode(sw1, INPUT_PULLUP);
13     pinMode(sw2, INPUT_PULLUP);
14     pinMode(sw3, INPUT_PULLUP);
15     pinMode(sw4, INPUT_PULLUP);
16     pinMode(sw5, INPUT_PULLUP);
17     pinMode(12, OUTPUT); // Green LED
18     pinMode(13, OUTPUT); // Red LED
19     lcd.begin(16, 2);
20     lcd.setCursor(0,0);
21     lcd.print("   Voting System");|
22     lcd.setCursor(0,1);
23     lcd.print("   Ready..");
24     delay(2000);
25     lcd.clear();
26 }
27 void loop() {
28     lcd.setCursor(0,0); lcd.print("RSP:"); lcd.print(votes[0]);
29     lcd.setCursor(8,0); lcd.print("UML:"); lcd.print(votes[1]);
30     lcd.setCursor(0,1); lcd.print("CON:"); lcd.print(votes[2]);
31     lcd.setCursor(8,1); lcd.print("Oth:"); lcd.print(votes[3]);
32     if (digitalRead(sw1) == LOW) registerVote(0);
33     if (digitalRead(sw2) == LOW) registerVote(1);
34     if (digitalRead(sw3) == LOW) registerVote(2);
35     if (digitalRead(sw4) == LOW) registerVote(3);
36     if (digitalRead(sw5) == LOW) showResult();
37 }
38 void registerVote(int candidate) {
39     votes[candidate]++;
40     digitalWrite(12, HIGH); // Green LED ON
41     delay(100);
42     digitalWrite(12, LOW);
43     while (digitalRead(sw1+candidate) == LOW); // Wait until released
44 }
45
```

```

45
46 void showResult() {
47     digitalWrite(13, HIGH); // Red LED ON
48
49     int maxVotes = 0;
50     int winner = -1;
51     bool tie = false;
52
53     for (int i = 0; i < 4; i++) {
54         if (votes[i] > maxVotes) {
55             maxVotes = votes[i];
56             winner = i;
57             tie = false;
58         } else if (votes[i] == maxVotes && maxVotes > 0) {
59             tie = true;
60         }
61     }
62     lcd.clear();
63     if (maxVotes == 0) {
64         lcd.print("No Voting Yet");
65     } else if (tie) {
66         lcd.print("Tie / No Result");
67     } else {
68         const char* names[4] = {"RSP", "UML", "CON", "Oth"};
69         lcd.print(names[winner]);
70         lcd.print(" Wins!");
71     }
72     delay(5000);
73     lcd.clear();
74     for (int i = 0; i < 4; i++) votes[i] = 0;
75     digitalWrite(12, LOW);
76     digitalWrite(13, LOW);
77 }
78

```

19. Light Intensity Measurement using LDR sensor: Circuit and code;



The image shows a circuit diagram and the corresponding Arduino code for measuring light intensity using an LDR sensor. The circuit includes an Arduino Uno, a 2.84V power source, a red LED, and an LDR sensor. The LDR sensor is connected to the Arduino's analog input pin A0. The red LED is connected to the Arduino's digital output pin 13. The code reads the sensor value and outputs it to the serial monitor.

Circuit Diagram:

- The Arduino Uno is connected to a 2.84V power source.
- A red LED is connected to the Arduino's digital output pin 13.
- An LDR sensor is connected to the Arduino's analog input pin A0.

Code:

```

1 // Analog sensor to PWM output
2
3 const int sensorPin = A0; // Analog input pin
4 const int pwmPin = 9; // PWM output pin
5
6 void setup() {
7     Serial.begin(9600); // Start serial monitor
8     pinMode(sensorPin, INPUT);
9     pinMode(pwmPin, OUTPUT);
10 }
11
12 void loop() {
13     // Read sensor value (0-1023)
14     int sensorValue = analogRead(sensorPin);
15
16     // Map to PWM range (0-255)
17     int pwmValue = map(sensorValue, 0, 1023, 0, 255);
18
19     // Output to PWM pin
20     analogWrite(pwmPin, pwmValue);
21
22     // Print values for debugging
23     Serial.print("Sensor: ");
24     Serial.print(sensorValue);
25     Serial.print(" PWM: ");
26     Serial.print(pwmValue);
27     Serial.println();
28     delay(100); // Small delay for stability
29 }
30

```

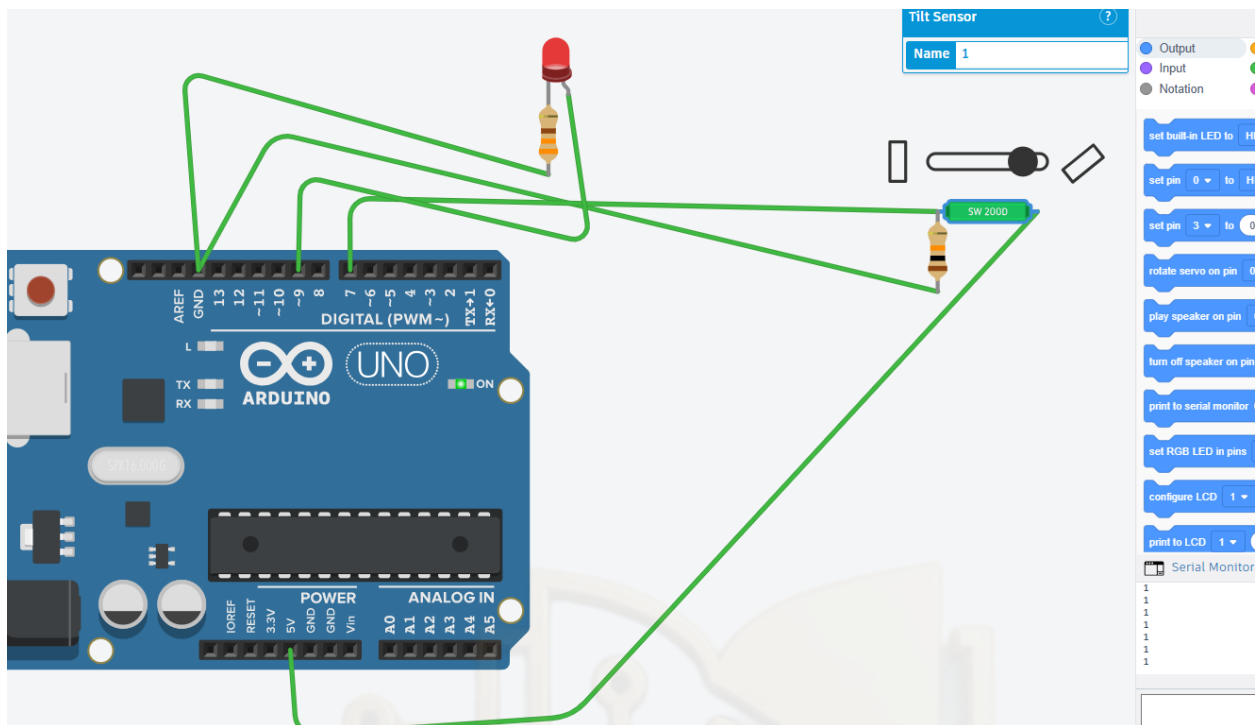
Serial Monitor Output:

```

Sensor: 687 PWM: 151
Sensor: 687 PWM: 151
Sensor: 687 PWM: 151
Sensor: 687 PWM: 151
Sensor: 687 PWM: 151
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Sensor: 687 PWM: 151
Sensor: 687 PWM: 151

```

20. Vibration sensor:



Code:

